Optical Communication Exam 2019: Annotated Version

Special instructions for invigilators: None.

Information for Candidates:

Numbers in brackets in the right margin (e.g. [5]) indicate maximum marks for each section of each question.

The following constants may be used:

electron charge : $e = 1.6 \times 10^{-19}$ C

permittivity of free space : $\varepsilon_o = 8.85 \times 10^{-12}$ F/m

relative permittivity of silicon : $\varepsilon_r = 12$

Planck's constant : $h = 6.63 \times 10^{-34}$ J s

Boltzmann's constant : $k = 1.38 \times 10^{-23}$ J/K

speed of light : $c = 3.0 \times 10^{8}$ m/s

The eigenvalue equations for TE modes in a symmetric slab waveguide of thickness d are

$\kappa = k_{1x} \tan(k_{1x} d/2)$ and $\kappa = -k_{1x} \cot(k_{1x} d/2)$

1. You should attempt all parts of this question. Short answers only are required; there is no need to re-state the questions in your answer book, but you should show any calculations you use to arrive at your answers, state any assumptions or approximations made, and give a brief (one or two lines) explanation where appropriate. All parts have equal value. [20]

a) An optical signal arrives at the end of a 50 km length of fibre with 1% of the intensity launched at the transmitter end. Calculate the fibre loss coefficient in dB/km.
A few people wrongly used the 0.99 reduction to calculate dB. The dB scale is 10log (Pout/Pin), so 10log(0.01), giving 20 dB loss.

b) A certain slab waveguide supports two TE modes, m=0 and m=1. For the m=1 mode, is the optical intensity higher at the centre of the guide, or at the core-cladding boundary? Give a brief explanation.
Some answers tried to use the fact that one mode is "bigger" than the other because less strongly guided. But the solution comes simply from the fact that odd numbered modes have zero intensity at the guide centre.

c) A certain slab waveguide supports 20 TE modes at a nominal wavelength λ_o = 780 nm. For an increased wavelength of λ_o = 1350 nm, would you expect the guide to support fewer, more or the same number of TE modes? Give a brief explanation.

d) If at some instant in time a 600 m length of optical fibre contains 3750 bits of propagating information, what is the bit rate of this signal?
Remember that the signal propagates at c/n', not c, and n' can be approximated as 1.5.

e) Why is silicon not a suitable material for constructing detectors for optical systems operating at nominal wavelength 1550 nm?
This does not relate to Si being indirect bandgap, that only affects light generation. It's just the bandgap being too large compared to the photon energy.

f) Calculate the slope efficiency of a laser diode operating at λ_o = 1330 nm if the quantum efficiency is 0.88.
Some people confused slope efficiency with responsivity (of a detector) – one is the inverse of the other (apart from the quantum efficiency).

g) If thermal noise and shot noise are equal in a certain optical fibre link, and then the fibre length is increased by 50%, which of these noise sources will now be greatest at the detector?

h) Estimate the spectral width in nm of an LED operating at a nominal wavelength of 580 nm.

i) A silicon p-n photodiode has p and n doping levels of $N_A = N_D = 4 \times 10^{20}$ m^{-3}. Calculate the bias voltage for which the maximum electric field strength will be 10^5 V/m.

j) A certain glass, for a nominal wavelength of λ_o = 600 nm, has a refractive index of $n = 1.50 + j \cdot 10^{-6}$. Find the attenuation coefficient at this wavelength, in units of m^{-1}.
This was intentionally a difficult one that few people got – see model solutions.

2. On pg. 1 the eigenvalue equations are given for TE modes in a symmetric slab waveguide as shown in Fig. 2.1 For a slab thickness d = 4 μm, a nominal wavelength $\lambda_o = 0.92$ μm, and refractive indices n_1 = 1.51 and n_2 = 1.50, and considering only TE modes:

a) How many modes does the guide support? [4]
Easy!

b) On a single plot, labelled appropriately, sketch the cross sectional shape $|E|^2$ vs x for each of the supported modes. [4]
Most people managed something, but often the shapes were poorly drawn, e.g. the m=1 mode not quite zero at the centre, or going to a sharp point at the centre when it should be smooth. Also be sure to label axes and note where the guide edges are.

c) Calculate the effective index n' for each of the supported modes to 3 decimal places. [8]

d) For a TE mode of order m=1 in an arbitrary symmetric slab waveguide, derive an expression for the position x_m at which the field intensity $|E|^2$ is maximised, as a function of the effective index n' of the mode, the nominal wavelength, and the core index n_1. [4]

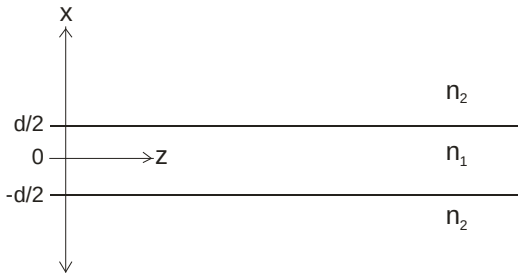


Figure 2.1 Slab waveguide

In questions like this make sure you read the question carefully and answer exactly what was asked.

3. a) How are group and phase velocity each defined for a propagating electromagnetic wave? What is the main physical significance of group velocity? [6]

b) In the wavelength range of interest, the refractive index of a sample of silica glass can be approximated by

$$n(\lambda_o) \cong D_o + D_1 \lambda_o^{-2} - D_2 \lambda_o^{2} \qquad (5.1)$$

where D_o = 1.45, D_1 = 0.003 μm^2 and D_2 = 0.0032 μm^{-2}. Find expressions for the phase and group velocities v_p and v_g in this material, as functions of D_o, D_1, D_2, and λ_o and c. Hence show that v_g is always $< v_p$ in this case. [6]

Main issue here tended to be messy and hard-to-follow derivations, which does lose you marks!

c) For the glass described above, find the wavelength of zero material dispersion. [4]

d) For the glass described above, derive an expression for the phase change $\Delta\varphi$ experienced by a wave of nominal wavelength λ_o propagating a distance L. [4]

Many people had trouble with this. See model answers.

4. a) Describe and discuss the important attenuation mechanisms in optical fibres, and the influence these have in choice of operating wavelengths. Use diagrams and equations where appropriate. [6]

Surprising number of people included comments about dispersion. Dispersion and attenuation are two completely different things!

b) An optical receiver can be implemented as a photodiode in series with a resistor, followed by a voltage amplifier. State the principal noise sources in this case, and draw a noise equivalent circuit for the receiver. [4]

Not many people did a good noise equivalent circuit diagram, although this can be found in the handout notes.

c) A certain optical link has a receiver with noise equivalent power (NEP) of 8.0 pW/√Hz, at the end of 100 km of fibre having attenuation of 0.35 dB/km at the operating wavelength of 1310 nm. The transmitted power is 4 mW and the photodetector quantum efficiency can be approximated as 1.

i) Determine whether shot noise or receiver noise is greater in this case; [4]

ii) Considering only the dominant noise source as determined in (i), find the maximum bit rate for an optical SNR of 12; [4]

iii) For the bit rate determined in (ii), find an approximate value for the maximum dispersion coefficient for this fibre above which the dispersion penalty would exceed 3 dB. Assume a transmitter spectral width σ_λ of 1.0 nm. [2]

Generally this was done quite well. Many people chose too large a dispersion time for iii).

5. a) A silicon p-n photodiode (Fig. 6.1) has a depletion layer thickness of w, and p and n doping levels respectively of N_A and N_D, with $N_A = 5N_D = 5\times10^{20}$ m^{-3}. The quantities w_p and w_n are the depleted widths in the p and n regions respectively. A reverse bias voltage V_b is applied. Find an expression for the full depletion width w as a function of V_b, and the value of V_b for which $w = 6$ μm. [8]

b) Neglecting Fresnel reflection, find an expression for the quantum efficiency η of the photodiode of (a) if the total p region thickness $h = 6$ μm, and the absorption coefficient $\alpha = 0.85\times10^5$ m^{-1}. Hence find the value of V_b for which $\eta = 0.80$. [6]

c) Give a typical approximate value for the saturation drift velocity of electrons in a semiconductor. For the photodiode of part (a) and an electron mobility in the silicon of 0.12 m^2/Vs, find the bias voltage for which the electron drift velocity is above this saturation value in half of the depleted region. [6]

Nothing particular to note on this question.